

MET CS662 - Assignment #7

1. Construct an npda's that accept the language $L = \{\omega \mid n_a(\omega) = n_b(\omega) + 1\}$ on $\Sigma = \{a, b, c\}$,

$$\begin{aligned} M &= (Q, \Sigma, \Gamma, \delta, q_0, Z, F) \\ Q &= \{q_0, q_1, q_2\} \\ \Sigma &= \{a, b, c\} \\ \Gamma &= \{A, B, Z\} \\ F &= \{q_2\} \\ \delta(q_0, \lambda, Z) &= \{(q_1, B)\} \\ \delta(q_1, a, Z) &= \{(q_1, AZ)\} \\ \delta(q_1, a, A) &= \{(q_1, AA)\} \\ \delta(q_1, a, B) &= \{(q_1, \lambda)\} \\ \delta(q_1, b, Z) &= \{(q_1, BZ)\} \\ \delta(q_1, b, A) &= \{(q_1, \lambda)\} \\ \delta(q_1, b, B) &= \{(q_1, BB)\} \\ \delta(q_1, c, Z) &= \{(q_1, Z)\} \\ \delta(q_1, c, A) &= \{(q_1, A)\} \\ \delta(q_1, c, B) &= \{(q_1, B)\} \\ \delta(q_1, \lambda, Z) &= \{(q_2, \lambda)\} \end{aligned}$$

2. Show that $L = \{a^n b^{2n} \mid n \geq 1\}$ is a deterministic context-free language.

$$\begin{aligned} M &= (Q, \Sigma, \Gamma, \delta, q_0, Z, F) \\ Q &= \{q_0, q_1, q_2\} \\ \Sigma &= \{a, b\} \\ \Gamma &= \{A, Z\} \\ \delta(q_0, a, Z) &= \{(q_0, AAZ)\} \\ \delta(q_0, a, A) &= \{(q_0, AAA)\} \\ \delta(q_0, b, A) &= \{(q_1, \lambda)\} \\ \delta(q_1, b, A) &= \{(q_1, \lambda)\} \\ \delta(q_1, \lambda, Z) &= \{(q_2, \lambda)\} \\ F &= \{q_2\} \end{aligned}$$

Since $L(M)$, and the following conditions are satisfied:

- For any $q \in Q, a \in \Sigma \cup \{\lambda\}, x \in \Gamma, |\delta(q, a, x)| \leq 1$.
- For any $q \in Q, x \in \Gamma$, if $\delta(q, \lambda, x) \neq \emptyset$, then $\delta(q, a, x) = \emptyset$ for every $a \in \Sigma$.

M is a deterministic pushdown automaton that recognizes L . Therefore, L is a deterministic context-free language.

3. Is the language $L = \{\omega c \omega^R \mid \omega \in \{a, b\}^*\}$ deterministic?

$$\begin{aligned}
M &= (Q, \Sigma, \Gamma, \delta, q_0, Z, F) \\
Q &= \{q_0, q_1, q_2\} \\
\Sigma &= \{a, b, c\} \\
\Gamma &= \{A, B, Z\} \\
\delta(q_0, a, Z) &= \{(q_0, AZ)\} \\
\delta(q_0, a, A) &= \{(q_0, AA)\} \\
\delta(q_0, a, B) &= \{(q_0, AB)\} \\
\delta(q_0, b, Z) &= \{(q_0, BZ)\} \\
\delta(q_0, b, A) &= \{(q_0, BA)\} \\
\delta(q_0, b, B) &= \{(q_0, BB)\} \\
\delta(q_0, c, Z) &= \{(q_1, Z)\} \\
\delta(q_0, c, A) &= \{(q_1, A)\} \\
\delta(q_0, c, B) &= \{(q_1, B)\} \\
\delta(q_1, a, A) &= \{(q_1, \lambda)\} \\
\delta(q_1, b, B) &= \{(q_1, \lambda)\} \\
\delta(q_1, \lambda, Z) &= \{(q_2, \lambda)\} \\
F &= \{q_2\}
\end{aligned}$$

Since $L(M)$, and the following conditions are satisfied:

- For any $q \in Q, a \in \Sigma \cup \{\lambda\}, x \in \Gamma, |\delta(q, a, x)| \leq 1$.
- For any $q \in Q, x \in \Gamma$, if $\delta(q, \lambda, x) \neq \emptyset$, then $\delta(q, a, x) = \emptyset$ for every $a \in \Sigma$.

M is a deterministic pushdown automaton that recognizes L . Therefore, L is a deterministic context-free language.

4. Give an example of a deterministic context-free language whose reverse is not deterministic.

$$L = \{a^n b^n \mid n \geq 1\} \cup \{b a^n b^{2n} \mid n \geq 1\}$$

L is a deterministic context-free language because there exists a deterministic pushdown automaton that recognizes the language. Two behaviors can be implemented depending on the first input symbol:

- If a is encountered first, then *one* symbol is pushed onto the stack for each a , then a symbol is popped for each b . If the stack becomes empty exactly as the b s finish, the string is accepted.
- If b is encountered first, then *two* symbols are pushed onto the stack for each a , then a symbol is popped for each b . If the stack becomes empty exactly as the b s finish, the string is accepted.

The first behavior is implemented with transition functions for the initial state q_0 , while the second behavior is implemented with transition functions for some other state q_1 .

When a b is encountered first, the transition function $\delta(q_0, b, Z) = \{(q_1, Z)\}$ is used to change states, thereby implementing the second behavior.

L^R is a non-deterministic context-free language because a pushdown automaton for L^R would always read b first. At this point, it needs to decide how to use the states:

- Should it stay at the same state and push a symbol for each b , popping a symbol for each a ?
- Should it move between two states and push a symbol for every other b , popping a symbol for each a ?

A deterministic PDA cannot make this choice. This requires nondeterminism – the PDA would need to guess which pattern to follow. Therefore, L^R is context-free but not deterministic context-free.

5. Show that the language $L = \{a^n b^j \mid n \leq j^2\}$ on $\Sigma = \{a, b, c\}$ is not context-free.

Use the Pumping Lemma for Context-Free Languages.

- Assume L is context-free
- If L is context-free, then there exists some integer $m \geq 1$ such that any string $w \in L$ with $|w| \geq m$ can be divided into five substrings $w = uvxyz$, satisfying the following conditions:
 - $|vxy| \leq m$
 - $|vy| \geq 1$
 - For all integers $i \geq 0$, the string $w_i = uv^i xy^i z \in L$
- Consider the string $w = a^{m^2} b^m$.
 - $w \in L$ because $n \leq j^2$: $m^2 = m^2$
 - $|w| \geq m$ because $|w| = m^2 + m$ and $m \geq 1$, so $m^2 + m \geq m$
- Decompose the string $w = a^{m^2} b^m$ so that vxy consists entirely of b .
 $v = b^k, y = b^l$, where $k + l \geq 1$ and $|vxy| \leq m$
- Pump using $i = 0$. The string $w_0 = a^{m^2} b^{m-(k+l)}$. Since $n \leq j^2$, it must be true that $m^2 \leq (m - (k + l))^2$. Some algebra follows:

$$m^2 \leq m^2 - 2m(k + l) + (k + l)^2$$

$$0 \leq -2m(k + l) + (k + l)^2$$

$$2m(k + l) \leq (k + l)^2$$

Since $k + l \geq 1$, it is possible to divide by $(k + l)$

$$2m \leq k + l$$

However, it was established that $|vxy| \leq m$, therefore $k + l \leq m$. This results in a contradiction because $m \geq 1$ and $k + l \geq 1$.

Therefore, $w_0 \notin L$ and L is not context-free.

6. Consider the language $L = \{a^n b^n c^m \mid n \geq 0, m \geq 0\}$. Show that this language is linear.

The language is linear because it can be generated by the linear grammar:

$$S \rightarrow Sc|A$$

$$A \rightarrow aAb|\lambda$$

7. Show that the family of linear languages is not closed under intersections

$$L_1 = \{a^n b^n c^m | n \geq 0, m \geq 0\} \text{ and } L_2 = \{a^k b^l c^l | k \geq 0, l \geq 0\}.$$

L_1 is a linear language as proved in the previous question. L_2 is a linear language because it can be generated by the linear grammar:

$$S \rightarrow aS|A$$

$$A \rightarrow bAc|\lambda$$

$$L = L_1 \cap L_2 = \{a^n b^n c^n | n \geq 0\}$$

The language $L = \{a^n b^n c^n | n \geq 0\}$ is a classic example of a language that is not context-free. The proof using the pumping lemma was demonstrated by the professor during the last lecture, so I will assume it be true. Since all linear languages are a subset of context-free languages, if L is not context-free, it cannot be linear.
